
SzCORE as a benchmark: report from the seizure detection challenge at the 2025 AI in Epilepsy and Neurological Disorders Conference

Jonathan Dan¹ Amirhossein Shahbazinia¹ Christodoulos Kechris¹ David Atienza¹

¹ Embedded Systems Laboratory, EPFL, Lausanne, Switzerland
jonathan.dan@epfl.ch

Abstract

Reliable automatic seizure detection from long-term electroencephalogram recordings (EEG) remains an open challenge in machine learning as current models often fail to generalize across patients or clinical settings. Manual EEG review remains the standard of care, highlighting the need for robust models and standardized evaluation. To rigorously assess current machine learning algorithms for automatic seizure detection in long-term EEG from the epilepsy monitoring unit, we organized a challenge. A private dataset of continuous EEG recordings from 65 subjects, totalling 4,360 hours of data, was utilized to evaluate algorithm performance. Expert neurophysiologists annotated these recordings, establishing the ground truth for seizure events. Algorithms were required to generate accurate onset and duration annotations, with performance measured using event-based metrics, including sensitivity, precision, F1-score, and false positive rate per day. The SzCORE framework was employed to ensure standardized evaluation across submissions. The event-based F1-score served as the primary ranking criterion, reflecting the clinical significance of detecting seizures while minimizing false positives. The challenge attracted 30 submissions from 19 teams or individuals, with 28 algorithms successfully evaluated. Results revealed significant performance variability among state-of-the-art approaches, with the top F1 score of 43% (sensitivity 37%, precision 45%), highlighting the persistent difficulty of this task for current machine learning methodologies. This independent evaluation also exposed a notable gap between self-reported efficacies and challenge performance, underscoring the critical need for standardized, rigorous benchmarking in developing clinically viable ML models. A comparison with previous challenges and commercial systems indicates that the best algorithm in this contest surpassed prior methods. Critically, the challenge infrastructure transitions into a continuously open benchmarking platform, fostering reproducible research and accelerating the development of robust seizure detection algorithms by allowing ongoing submissions and integration of additional private datasets. Clinical centres can also adopt this platform to evaluate seizure detection algorithms on their EEG data using a standardized, reproducible framework.

1 Introduction

Automated seizure detection is an active and critical area of machine learning research, particularly driven by the increasing use of ambulatory and long-term electroencephalography (EEG) monitoring in patients with epilepsy. These monitoring techniques, such as those used in epilepsy monitoring units (EMU) and home-based EEG, aim to provide continuous, real-time insights into a patient's condition, improving diagnostic accuracy and enabling timely medical interventions. Real-time

seizure detection can significantly benefit clinical practice by providing immediate alerts to healthcare professionals during in-hospital monitoring Kamitaki et al. [2019]. Meanwhile, offline detection helps reduce clinicians’ workload and detect subtle, often difficult-to-identify seizures Baumgartner and Koren [2018].

Despite significant advancements in the development of automated methods, algorithms often struggle to generalize across different patients, clinical settings, and seizure types. As a result, manual EEG review by clinicians remains the gold standard, underscoring the urgent need for robust, generalizable algorithms that can be deployed in real-world clinical environments. This gap in the state of the art presents a significant opportunity for innovation in machine learning and medical applications.

One critical obstacle hindering progress in this field is the lack of standardized evaluation practices, which complicates direct comparisons across studies and prevents the community from identifying the best-performing algorithms. Many recent reviews of seizure detection methods provide valuable overviews of existing work but fail to compare state-of-the-art seizure detection algorithms directly. Miltiadous et al. [2023] conducted a systematic review according to the PRISMA guidelines. They report the performance metrics reported by the authors for 89 studies in the Bonn dataset Andrzejak et al. [2001], 20 studies in the Physionet CHB-MIT Scalp EEG database Goldberger et al. [2000], Shoeb [2009], 16 in other datasets, and 65 in combinations of datasets. The results are organized by dataset. Shoebi et al. [2021] focused on categorizing deep learning methods. They report the accuracy of the authors’ findings for 136 studies grouped by deep-learning technique. Similar approaches have been developed in other recent reviews Supriya et al. [2023], Siddiqui et al. [2020], Baumgartner and Koren [2018]. Although these reviews provide a comprehensive overview of the recently published work with a detailed analysis of the datasets used for training and evaluation, along with categorization of the most common machine learning techniques, they do not allow a direct comparison of performance due to heterogeneity in the data sets, evaluation metrics, and evaluation methodology. Most reviews acknowledge this limitation.

To address these challenges, Dan et al. developed SzCORE Dan et al. [2024], a framework that standardizes the validation methodology of seizure detection algorithms and enables direct comparison across studies. SzCORE proposes guidelines on EEG datasets, evaluation methodologies, and performance metrics. It defines the file format and data organization of EEG datasets. These follow a specific format that complies with the BIDS-EEG Gorgolewski et al. [2016], Pernet et al. [2019] and HED-SCORE Hermes et al. [2024] specifications. SzCORE defines event-based scoring to compute performance metrics. Event-based scoring compares the labels of an event (seizure) based on the overlap between the reference (ground truth) and the hypothesis (algorithm output) to count True Positives (TP), False Positives (FP) and False Negatives (FN). These counts then compute four performance metrics: sensitivity, precision, F1-score, and false positive rate per day (FP/24h). Algorithms evaluated with SzCORE can be directly compared as performance metrics are evaluated using the same methodology. By providing this framework, SzCORE aims to accelerate iterative improvements in machine learning models for seizure detection, allowing researchers to more reliably build upon prior art.

A standardized evaluation framework, such as SzCORE, provides a consistent methodology for comparing algorithms, making it an ideal basis for self-evaluation. However, it does not provide a submission and evaluation pipeline for benchmarking algorithms on a private dataset.

This paper presents the results of the seizure detection challenge we organized as part of the 2025 International Conference on Artificial Intelligence in Epilepsy and Neurological Disorders held in Breckenridge (Colorado, USA) from March 3 to 6, 2025. The challenge aimed to identify the best algorithms for automated seizure detection based on long-term scalp EEG from the EMU. The SzCORE framework was used for a standardized evaluation. This paper presents the first large-scale direct comparison of seizure detection algorithms using a standardized evaluation methodology on a substantial private dataset. Furthermore, by analyzing the characteristics of submitted algorithms alongside their performance, this work aims to shed light on effective strategies and persistent hurdles in developing automated seizure detection systems. The challenge’s source code and evaluation platform will remain online as a public service.

This initiative provides a crucial and persistent resource for the machine learning community, promoting transparency, reproducibility, and continuous progress in this vital area of medical AI, opening up a new way of benchmarking seizure detection algorithms on private datasets.

We describe the organization and timeline of the challenge in Section 2, the dataset (Section 3), the evaluation procedure (Section 4) and the algorithmic contributions (Section 5). We analyze the submission results in terms of event-based F1-score and compute cost (Section 6). We compare them to self-reported results, previous challenges, and commercial software and discuss the limitations of the challenge in the discussion (Section 7).

2 Challenge description

This challenge aimed to build a seizure detection algorithm that could detect the onset and duration of all epileptic seizures in a long-term (multi-day) EEG recording from the EMU. Long-term continuous EEG signals from the EMU were provided as input data. The recordings were stored in .EDF files. EEG acquisition comprised 19 electrodes in the international 10-20 system in a referential, common average montage ^{??}. The channels were provided in the following order: Fp1-Avg, F3-Avg, C3-Avg, P3-Avg, O1-Avg, F7-Avg, T3-Avg, T5-Avg, Fz-Avg, Cz-Avg, Pz-Avg, Fp2-Avg, F4-Avg, C4-Avg, P4-Avg, O2-Avg, F8-Avg, T4-Avg, T6-Avg (see Section 3 for details on standardization). The recordings were sampled at 256 Hz. Each file was guaranteed to last at least 1 minute, with most files lasting approximately an hour.

The algorithms generated a tab-separated value .TSV file as an output. This text file uses a tab as a delimiter to separate the different columns of information, each row representing one seizure event. Each annotation file is associated with a single EEG recording. The annotation file was required to be HED-SCORE compliant Dan et al. [2024]. It contains the onset and duration of each event.

The participating teams were asked to submit a pre-trained algorithm packaged as a Docker image. The participants also provided an abstract describing their method and a list of datasets used for training. The algorithm could be trained using any combination of public and private datasets. Participants were provided SzCORE formatted versions of the Physionet CHB-MIT Scalp EEG Database, the Physionet Siena Scalp EEG Database, and the TUH EEG Sz Corpus to facilitate development and ensure a common baseline for publicly available data. Although open-source submissions were encouraged, our Docker-based approach allowed for closed-source participation.

The challenge ran from 1 December 2024 to 16 February 2025; the results were announced on 4 March 2025.

3 Evaluation dataset

The evaluation dataset comprised previously collected recordings from 65 patients (resulting in 398 annotated seizures) in the EMU of the Philadelphia Danish Epilepsy Centre in Dianalund, between January 2018 and December 2020. Data had been recorded with the NicoletOne™ v44 amplifier. The data contained at least one seizure during the hospital stay with a visually identifiable electrographic correlate to clinical seizures recorded on video. The total recording duration amounted to 4360 hours. Subject recordings ranged from 18 hours to 98 hours. Most participants were adults, with a median age of 34 years (range: 5 to 66 years). Eight children were included in the study. Seizures had been independently annotated by three board-certified neurophysiologists with expertise in long-term video-EEG monitoring. In case of disagreement, a ground-truth label was obtained after a consensus discussion between the experts, ensuring high-fidelity ground truth for the evaluation. The statistics of the dataset are illustrated in Figure 1. Data were anonymized and then converted to a BIDS-compatible Gorgolewski et al. [2016], Pernet et al. [2019] version using the `epilepsy2bids` Python library. The library was adapted to support the Philadelphia dataset. It is publicly available on GitHub. The conversion standardizes the EEG channels’ number, naming, and order to the 19 channels of the 10-20 system. They are re-referenced to a common average montage and uniformly sampled at 256 Hz.

4 Evaluation procedure and metrics

The submission procedure required participants to submit pre-trained algorithms packaged as Docker images to ensure reproducibility, compatibility, and ease of execution across computing environments. Docker images were tested on a development server to verify compatibility with the required submission format, including valid annotation output. The Docker images were then

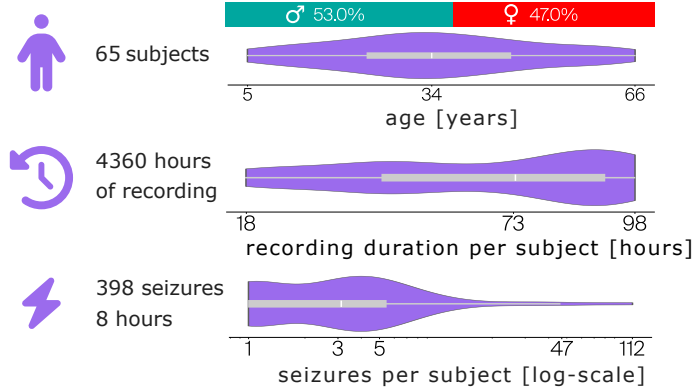


Figure 1: Distribution of the data in the Filadelfia dataset.

executed in an isolated environment on an institutional computing platform to compute seizure annotations for all recording files in the evaluation dataset (see Section 3). The performance metrics were then calculated by comparing the algorithm annotations with the ground truth annotations.

Institutional computing platform provided access to a GPU cluster through a Container-as-a-Service (CaaS) model, enabling efficient and flexible use of more than 400+ GPUs of different types. In the context of this challenge, A100 NVIDIA GPU nodes with 2 x CPU AMD EPYC 7543 32c/64t, 1 TB RAM, and 8 x NVIDIA HGX A100-SXM4-80GB NVLINK were used for GPU workloads. V100 NVIDIA GPU nodes with 2x INTEL Gold 6240, 384 GB RAM and 4x NVIDIA v100-SXM2-32GB NVLINK were used for CPU-only loads. The computing platform provides detailed reports on the total GPU hours, CPU hours, and average RAM usage per hour used to execute a compute job. The GPU hours are calculated based on reservation time, whereas CPU and RAM usage are based on actual utilization. We manually determined the execution parameters by evaluating algorithms on a small subset of the dataset; a pragmatic approach given the diversity of submitted computational requirements. This determined the GPU reservation requirements and the number of files to process in parallel within each Docker container. We further split the inference across multiple Docker instances when required to accelerate execution.

The performance metrics reported in the challenge include sensitivity, precision, F1-score, and false positive per day. They are computed using the `timescoring` library, which is described in SzCORE Dan et al. [2024]. The library provides both sample-based and event-based scoring. In this challenge, event-based scoring was selected as the primary evaluation mode for algorithm ranking because it better reflects clinical decision-making. This scoring methodology focuses on each seizure as a single event, thereby providing an accurate count of the number of detected events and false alarms the detection system would trigger. Event-based scoring relies on overlap: if a reference (ground-truth) event and a hypothesis (algorithm) event overlap, the detection is correct (True Positive); if a hypothesis event does not overlap with any reference event, it is considered a false detection (False Positive). The following event-based scoring parameters (default values from the SzCORE framework) are used in this challenge:

- Minimum overlap: between the reference and hypothesis for a detection. We use any overlap, however short, to enhance sensitivity.
- Pre-ictal tolerance: tolerance with respect to the onset of an event that would count as a detection. We use a 30 seconds pre-ictal tolerance.
- Post-ictal tolerance: tolerance with respect to the end time of an event that would still count as a detection. We use a 60 seconds post-ictal tolerance.
- Minimum duration: between events resulting in merging events that are separated by less than the given duration. We merge events separated by less than 90 seconds which corresponds to the combined pre- and post-ictal tolerance.
- Maximum event duration: splitting events longer than the given duration into multiple events: We split events longer than 5 minutes.

Scores are computed per subject by adding the true positives, false positives and false negatives of the individual recordings to calculate sensitivity, precision, F1-score and false positives per day. These scores are then averaged with the arithmetic mean over the different subjects. The `szcore-evaluation` library was used to compute performance metrics across the entire dataset. In case an algorithm does not produce a .TSV seizure annotation file for a given .EDF file, it is considered that the algorithm did not predict any seizures in that file.

The primary ranking criterion was the event-based F1-score, as it balances seizure detection rate (sensitivity) against precision. It is computed as the harmonic mean of sensitivity and precision. This metric represents the predictive performance of an algorithm. It is particularly relevant in the context of epileptic seizure detection, where the primary objective is the detection of seizures, which are rare events.

5 Algorithm submissions

Nineteen participants took part in the challenge. This resulted in 30 submissions, leading to 20 distinct algorithms after considering unique approaches and minor variations. Submission methods followed a popular scheme in which the algorithm processes the input signal in fixed-duration windows. The window then slides with a predefined step to the next input. The window duration ranged from a few seconds to a few minutes. The algorithm processing can typically be described in three steps: 1. pre-processing the input windows - filtering and sometimes extracting signal features, 2. processing the inputs with a machine-learning model, 3. post-processing the model outputs.

Pre-processing Most algorithms included a pre-processing step comprising a band-stop and / or band-pass filter Sanei and Chambers [2007]. Some algorithms extracted features of the EEG after filtering. Four algorithms used the short-time Fourier transform (STFT) of the EEG as input to the model. The *Gradient Boost v1*, *v2*, *v3* algorithms extracted linear and non-linear connectivity features. The *Random Forest* algorithm combined time features with frequency ones. The *NE Illusion* algorithm included a channel selection step. Similarly, the *STORM* algorithm removed low-quality EEG channels.

Processing Deep learning methods were used in 17 submissions, with 13 algorithms processing inputs in the time domain, while the four others processed STFT inputs.

Post-processing included several variations to smooth the model output. Implementations included moving average or median filters, merging nearby seizure events, discarding short ones, and thresholding predictions.

Participants were allowed to submit multiple versions of their method, resulting in five methods with multiple versions, ranging from two to three versions each. Variations between versions included:

1. Changes in the classification threshold observed in *Gradient Boost*, *HySEIZ* and *S4Seizure*.
2. Changes in the model architecture observed in *HySEIZ*.

The steps and methods used by each submission are detailed in Appendix A (Table 2). The relevant information for each method was extracted from the accompanying abstract, provided by the development team during the submission stage, and their corresponding GitHub repository. The information was verified with the algorithm developers. The datasets used to develop each train are detailed in Appendix B (Table 3).

6 Competition results

Twenty-eight of the thirty algorithm submissions were evaluated. Two of the submissions could not be evaluated. One contained an erroneous link to the Docker image. The other would crash upon execution due to errors in the packaging of the code libraries. The event-based scores are presented in Table 1. The winning submission (*Sz Transformer*) has an F1-score of 43%, a sensitivity of 37%, a precision of 45%, and produces an average of 1.34 false positives per day. The top-5 submissions have a sensitivity ranging from 37%-58%, a precision of 27%-45% and produce between 1.34 and

14 false positives per day. The sensitivity in function of precision is shown for all submissions in Figure 2..

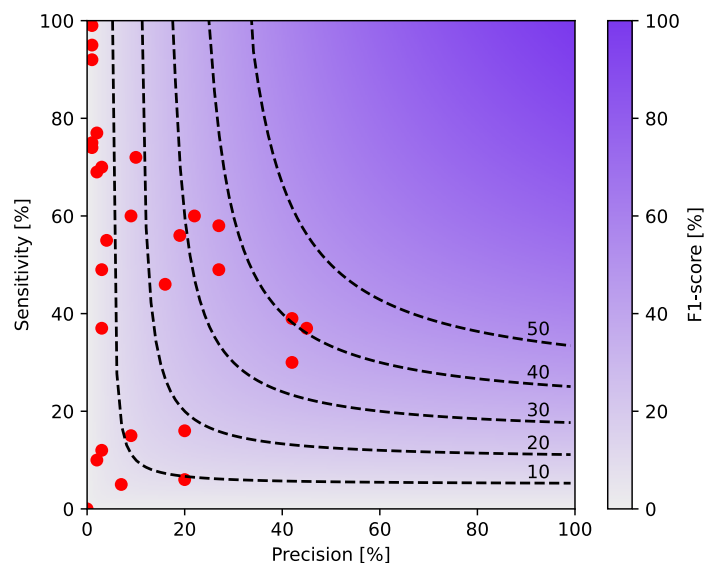


Figure 2: Sensitivity as a function of precision for the • algorithms submitted in the challenge. The background is shaded according to the F1-score, with dashed lines indicating iso-F1 score.

The agreement between the algorithms is shown in Figure 3. It shows that a single seizure event was missed by all algorithms, and that the seizure with the most detections was detected by 26 out of the 28 algorithms. Notably, the figure also shows low agreement between false positives, with less than 25% of the algorithms agreeing on 91% of the false detections. This suggests that false positives are largely idiosyncratic to individual algorithms rather than stemming from universally misleading artifacts in the data.

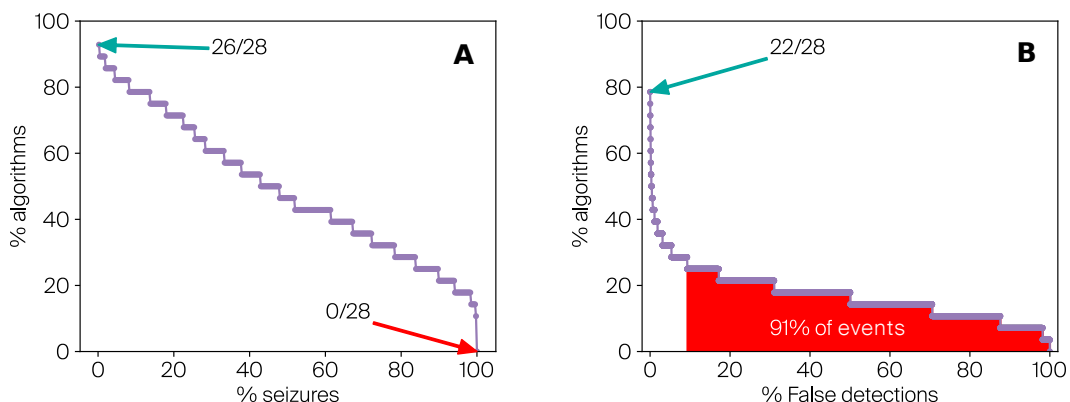


Figure 3: Percentage of algorithms that detect a percentage of events. Panel **A** shows the true positives, and panel **B** shows the false positives.

The difference between the performance reported by algorithm developers on their self-selected datasets and the event-based F1-score on the challenge evaluation dataset is represented in Figure 4. This comparison starkly illustrates that nearly all algorithms strongly overestimated their performance compared to the results obtained on this independent test set, highlighting a critical challenge in the field regarding generalizability and the necessity of standardized, independent benchmarking.

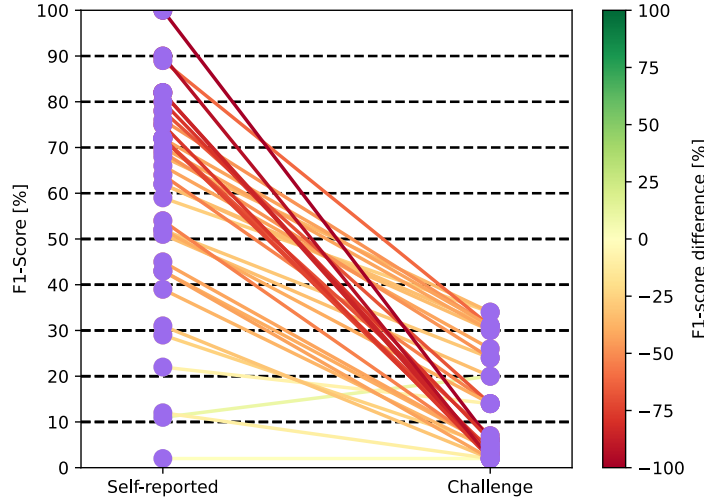


Figure 4: F1-score self-reported by the algorithm developers versus the event-based F1-score obtained in this challenge. The difference in F1-score is represented by the color of the line.

Evaluating all the algorithms required 334 Nvidia A100 GPU hours, 97’885 CPU hours (76’668 hours of Intel Gold 6240 and 21’217 hours of AMD EPYC 7543 32c/64t) and 403’933 GB of RAM. The computing resources are detailed in Appendix C (Table 4).

7 Discussion

This challenge aimed to rigorously evaluate the current state of machine learning algorithms for automatic seizure detection in long-term EEG recordings from the EMU, leveraging a large, private dataset and the standardized SzCORE evaluation framework. With 30 submissions from 19 teams, this initiative represents a significant, contemporary cross-section of methodologies in the field.

The organization of challenges is a recognized catalyst for innovation within machine learning, famously demonstrated by events like the ImageNet Large-Scale Visual Recognition Challenge Rusakovsky et al. [2015], which was pivotal in showcasing the power of Convolutional Neural Networks. Within epilepsy research, previous challenges have targeted diverse aspects, from intracranial EEG up to [2014] to optimizing electrode configurations neu [2020] and wearable EEG Chatzichristos et al. [2023]. Our challenge builds upon these efforts, uniquely focusing on the prevalent 19-channel scalp EEG setup from EMUs. The scale of participation (28 successfully evaluated algorithms) and the performance of the top submissions (F1-score of 43%, sensitivity 37%, precision 45% at 1.34 FP/day) indicate an advancement over prior scalp-EEG challenges. For instance, the leading algorithm in the Neureka challenge achieved 12.37% sensitivity at 1.44 FP/day on a different dataset and scoring methodology Shah et al. [2018], suggesting tangible progress in algorithmic capabilities.

When benchmarked against commercial systems as evaluated by Koren et al. Koren et al. [2021] (who used a comparable, albeit distinct, dataset and slightly more permissive event-scoring), our top-performing algorithms demonstrate competitive F1-scores (e.g., Encevis 1.7 achieved 44.1%). This suggests that research-driven algorithms are approaching, and in some aspects matching, the performance of established commercial solutions. However, it’s crucial to note that commercial systems often operate at a different point on the sensitivity-precision curve, typically prioritizing higher sensitivity (68-81%) at the cost of increased false positives, a trade-off reflecting specific clinical needs.

The choice of the event-based F1-score as the primary ranking criterion in this challenge encouraged participants to find a balance between sensitivity and precision. Our results, particularly the analysis of multiple submissions from the same team with varied thresholds (e.g., *S4Seizure*), illustrate the

Algorithm	F1-score	Sensitivity	Precision	FP/day
Sz Transformer	43	37	45	1
Van Gogh Detect	36	39	42	3
S4Seizure v2	34	30	42	2
DeepSOZ-HEM	31	58	27	14
S4Seizure v1	30	49	27	7
HySEIZa v1	26	60	22	13
S4Seizure v3	24	56	19	13
zhu-transformer	20	46	16	24
SeizUnet	19	16	20	4
HySEIZa v2	14	72	10	29
Channel-adaptive	14	6	20	1
eventNet	14	60	9	20
Gradient Boost v2	7	15	9	6
DynaSD	6	55	4	37
Random Forest	6	5	7	1
SD2025 v2	5	70	3	86
bRAWIstar	5	49	3	46
Gradient Boost v1	4	12	3	12
SeizFormer	4	77	2	83
Band Powers	4	37	3	50
NE Illusion 1 v1	3	69	2	158
Gradient Boost v3	3	10	2	22
NE Illusion 1 v3	2	95	1	280
STORM	2	99	1	290
eegwavenet	2	92	1	237
ConvNet	2	75	1	163
SD2025	2	74	1	188
NE Illusion 1 v2		0		0

Table 1: Event-based scores of the competition submissions.

critical impact of operating point selection. However, the challenge format did not allow participants to fully explore the sensitivity-precision curve, which might limit the clinical utility of the winning solution. An important direction for future research is to develop methods that can robustly optimize this trade-off or allow clinicians to easily tune it post-hoc based on their specific requirements, without needing access to the original evaluation dataset.

A striking finding from this challenge is the significant discrepancy between the self-reported performance of algorithms on developers' chosen datasets and their actual performance on our independent evaluation dataset (Figure 4). This observation, common in machine learning but particularly critical in medical AI, underscores the profound impact of dataset characteristics and evaluation methodologies. The Philadelphia dataset, with its inclusion of pediatric patients and data from portable EEG systems allowing patient mobility, likely presents a more complex and realistic distribution of signals and artifacts than many curated public datasets. This "generalization gap" highlights the insufficiency of relying solely on self-reported metrics and the imperative need for independent, standardized benchmarking on diverse, clinically representative data. Future initiatives could further enhance generalizability assessment by incorporating multi-centric evaluation datasets, reflecting the wider variability in EMU equipment and protocols.

A core contribution of this work, extending beyond the immediate results of the challenge, is the transition of the entire evaluation infrastructure into a continuously open benchmarking platform. This initiative directly addresses the aforementioned critical need for standardized, reproducible, and ongoing assessment in the field of seizure detection. By allowing continuous submissions, this platform will serve as a living benchmark, enabling researchers to:

- Evaluate new algorithms against a consistent, private, and expertly annotated dataset under a unified scoring framework (SzCORE).
- Objectively track the progress of the field over time, mitigating the issue of self-reporting bias.

- Foster transparency and reproducibility, as methods can be directly compared.
- Potentially integrate additional private datasets from other clinical centers under the same standardized evaluation pipeline, broadening the benchmark’s scope and robustness.

This open platform is designed to become a vital community resource, significantly lowering the barrier to rigorous evaluation and accelerating the iterative development and validation of robust seizure detection algorithms. For the Neurips community, this provides a concrete tool and a new paradigm for benchmarking in a challenging medical domain, promoting best practices in ML evaluation.

Finally, while the top-performing algorithms in this challenge show promise, their direct translation to clinical practice requires further steps. The current F1-scores, while competitive, may still fall short of clinical expectations for sensitivity in many scenarios. Further algorithmic refinement is needed, potentially focusing on achieving much higher sensitivity, even if it incurs a higher false positive rate that clinicians might manage with appropriate alert systems and review interfaces. The standardization efforts inherent in SzCORE and this benchmarking platform can facilitate this transition by ensuring that new algorithms can be seamlessly integrated into EEG review software that adheres to these established input/output formats. This framework supports the development of more sophisticated clinical decision support tools, ultimately benefiting patient care and easing clinician workload.

8 Conclusion

The 2025 Seizure Detection Challenge provided a rigorous and standardized assessment of contemporary machine learning algorithms for seizure detection in long-term EMU scalp EEG, utilizing the SzCORE framework for transparent and comparable evaluation. While results demonstrated notable advancements and the competitiveness of leading algorithms against even commercial software, they also highlighted the persistent difficulties in achieving high sensitivity and precision simultaneously, and starkly revealed the generalization gap between self-reported efficacy and performance on unseen, diverse clinical data.

More significantly than the snapshot of current performance, this challenge transitions into a continuously open benchmarking platform. This initiative, a core outcome of our work, establishes a lasting resource for the machine learning and epilepsy research communities. By providing ongoing access to a standardized evaluation pipeline on a private, expertly annotated dataset, the platform directly addresses the critical need for reproducible research, independent validation, and continuous tracking of progress in the field. It offers a means to objectively compare novel approaches, iterate on existing models, and reduce the discrepancy often observed in self-reported results.

This open benchmark is designed to foster sustained innovation beyond the timeframe of a single competition. It will enable researchers to consistently test their algorithms, allow clinical centers to evaluate solutions on their own data using a standardized methodology, and collectively accelerate the development of robust, generalizable seizure detection models. Ultimately, by transforming this competitive event into a persistent community resource, we aim to catalyze the progress towards clinically impactful machine learning tools that can meaningfully improve patient care in epilepsy.

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A Algorithmic Details

Algorithm	Method	Input	Notch	Pre-processing			Features	Post-proc.	Reported F1 [%]	Code
				Bandpass	Resample					
bRAWIstar	Linear Recurrent Unit	–	60	✗	✗	✗	✗	–	29–45	✗
Channel-adaptive	CNN	✗	✗	✗	✗	✗	✗	✓	78	🏆
ConvNet	CNN	10	✗	1–45	✗	✗	✗	✗	80–90	✗
DeepSOZ-HEM	LSTM & Transformer	600	✗	✗	✗	✗	✗	✓	89	🏆
DynaSD	LSTM	1	60, 120	1–100	128 Hz	✗	✗	✓	✗	🏆
eegwavenet	EEGWavenet & CNN	4	✗	–64	✗	✗	✗	–	2–31	🏆
eventNet	U-Net & CNN	120	✗	✗	✗	✗	✗	✗	22–52	🏆
Gradient Boost	Gradient Boosted Trees	10	✗	✗	✗	✗	✓	✓	72	🏆
HySEIZa	Hyena-Hierarchy & CNN	12	50, 60	0.5–120	✗	✗	✓	✓	66.01	✗
NE Illusion 1	JEPA Assran et al. [2023]	5	60	0.5–124.9	✗	✗	✓	✗	82	✗
Random Forest	Random Forest	2	✗	1–50	✗	✗	✓	✓	80	✗
S4Seizure	S4 Gu et al. [2022]	12	50, 60	0.5–120	✗	✗	✓	✓	62–70	✗
SD2025	LaBraM Jiang et al. [2024]	5	50	✗	200 Hz	✗	✗	✗	43	✗
STORM	BERT Devlin et al. [2019]	60	–	1–70	–	–	–	–	75.93	✗
SeizFormer	CNN & Transformer	12	50, 60	0.5–129	–	–	✓	–	54.65	✗
SeizUnet	U-net & LSTM	20.48	✗	0.5–100	200 Hz	✗	✗	✓	60.6	🏆
Sz Transformer	CNN & Transformer	–	1, 60	0.5–120	✗	✗	✗	–	✗	🏆
Band Powers	–	5	50	2–40	–	–	✓	–	39	✗
Van Gogh Detect	CNN & Transformer	13x10	–	–	–	–	–	✓	60	✗
zhu-transformer	CNN & Transformer	25	–	–	–	–	–	–	11–51	🏆

Table 2: Algorithmic details for all submissions. Input duration is expressed in seconds. Post-processing involves variations of prediction output smoothing. The reported F1-score is evaluated by the methods’ authors on different validation sets, often hold-outs from the training set. Missing information is marked by a –.

B Training Datasets

Method	TUH Seizure	TUH EEG	Siena Scalp	CHB-MIT	SeizIT2	HUP	Total
bRAWIstar	✓	✗	✗	✗	✗	✗	1
Channel-adaptive	✓	✗	✓	✓	✗	✗	3
ConvNet	✗	✗	✗	✓	✗	✗	1
DeepSOZ-HEM	✓	✗	✓	✗	✗	✗	2
DynaSD	✗	✗	✗	✓	✗	✓	2
eegwavenet	✗	✗	✗	✓	✗	✗	1
eventNet	✓	✗	✗	✗	✗	✗	1
Gradient Boost	✓	✗	✓	✗	✗	✗	2
HySEIZa	✓	✗	✗	✗	✗	✗	1
NE Illusion 1	✓	✗	✗	✗	✗	✗	1
Random Forest	✗	✗	✗	✓	✗	✗	1
S4Seizure	✓	✗	✗	✗	✗	✗	1
SD2025	✓	✓	✓	✗	✗	✗	3
STORM	✓	✓	✗	✗	✗	✗	2
SeizFormer	✓	✗	✗	✗	✗	✗	1
SeizUnet	✓	✗	✗	✗	✓	✗	2
Sz Transformer	✓	✗	✓	✗	✗	✗	2
Band Powers	✗	✗	✓	✗	✗	✗	1
Van Gogh Detect	✓	✗	✗	✓	✗	✗	2
zhu-transformer	✓	✗	✗	✗	✗	✗	1
Total	15	2	6	6	1	1	

Table 3: Training datasets used by each method. The public datasets used were: TUH EEG Seizure Corpus Shah et al. [2018], TUH EEG corpus Obeid and Picone [2016], Siena Scalp EEG Databaset Detti [2020], Detti et al. [2020], Goldberger et al. [2000], Short-term SWEC iEEG Burrello et al. [2019], Physionet CHB-MIT Scalp EEG dataset Shoeb [2009], Goldberger et al. [2000], KU Leuven SeizIT2 Bhagubai et al. [2025], Intracranial EEG recordings of Seizures from HUP Bernabei and Litt [2021].

C Computing Resources

Algorithm	GPU [h]	CPU [h]	RAM [GB/h]
Sz Transformer	1	128	173
Van Gogh Detect	10	161	254
S4Seizure v2	10	1'168	430
DeepSOZ-HEM	1	44	74
S4Seizure v1	9	1'186	630
HySEIZa v1	4	72	153
S4Seizure v3	9	1'170	547
zhu-transformer	85	7'952	8'708
SeizUnet	14	210	510
HySEIZa v2	5	69	177
Channel-adaptive	25	82	122
eventNet	1	108	161
Gradient Boost v2	0	6'006	86'681
DynaSD	0	966	2'116
Random Forest	0	223	937
SD2025	3	172	146
bRAWIstar	3	203	559
Gradient Boost v1	0	6'913	85'688
SeizFormer	6	61	232
Band Powers	2	52	124
NE Illusion 1 v1	59	3'912	5'830
Gradient Boost v3	0	6'152	72'787
NE Illusion 1 v3	46	6'882	19'030
STORM	0	48'773	104'361
eegwavenet	1	107	177
ConvNet	2	3'494	12'234
SD2025 v2	5	188	377
NE Illusion 1 v2	31	1'431	715
Total	335	97'885	403'933

Table 4: Computing resources